



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

DISTRIBUTED LEDGER PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Blockchain/Web3

TOTAL: 10 QUESTIONS

Q1 What is Distributed Ledger and what AI/ML use case is it designed for? What Model

Q6 How do you evaluate a model's performance in Distributed Ledger? Distributed Ability

Q2 What are the core abstractions in Distributed Ledger? Architecture

Q7 How do you deploy a model trained with Distributed Ledger? Deployment Cycle

Q3 How does Distributed Ledger handle training or inference? Deployment

Q8 How does Distributed Ledger handle distributed or multi-GPU training? Deployment

Q4 How does Distributed Ledger manage data preprocessing? Deployment

Q9 How do you track experiments and versions in Distributed Ledger? Deployment

Q5 How does Distributed Ledger support GPU/TPU acceleration? Deployment

Q10 What are common pitfalls when using Distributed Ledger in production? Deployment



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Distributed Ledger is a framework or service

Mitigation

Distributed Ledger is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set and

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 Distributed Ledger organizes work around abstractions like

Sharding

Distributed Ledger organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A7 Export the model to a portable format

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, Distributed Ledger defines a model

Hardening

For training, Distributed Ledger defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 Distributed Ledger provides data-parallel or model-parallel

Performance

Distributed Ledger provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 Distributed Ledger provides data loaders, tokenizers, or Gotchas

Gotchas

Distributed Ledger provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights &

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside Distributed Ledger to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 Distributed Ledger detects available accelerators and moves

Migration

Distributed Ledger detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew (different

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.